Process Characterization Report

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCR-009

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

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Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; Cpk, process capability index (one-sided); CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SDM, scale-down model; SEC, size-exclusion chromatography; SOP, standard operating procedure; TCID50, fifty per cent tissue culture infectious dose; UF/DF, ultrafiltration and diafiltration; VF, virus filtration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Small-virus retentive filtration is the dedicated size-based virus removal step of the A-Mab drug-substance process, and it is the principal contributor to the minute virus of mice (MVM) clearance claim. This report covers the characterization executed under the paired plan PCP-009 on a qualified scale-down spiking model, over the 2 operating parameters that the pre-characterization risk assessment (RA-001) carried into multivariate study. The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development principles of ICH Q8, ICH Q9 and ICH Q11. The clearance claim is framed by ICH Q5A(R2) (International Council for Harmonisation 2023). The process model, the quality attribute framework and the modular clearance approach are those of the A-Mab case study (CMC Biotech Working Group 2009).

MVM clearance falls as the volumetric load on the membrane rises. Across the characterized load range of 50–140 L/m² the response-surface model predicts a decrease of 1.76 $\log_{10}$, and volumetric load is the only parameter whose effect on MVM clearance was resolved ($p < 0.001$). Each additional 10 L/m² of load costs about 0.20 $\log_{10}$ of retention near the set-point. The trend is bounded by the acceptance criterion for the step. At the worst corner of the characterized region, which is the highest load at the highest pressure, the predicted MVM log reduction is 4.48 against a required step contribution of 3.89 $\log_{10}$, so the criterion is met throughout the region studied (§6). Within the normal operating ranges the predicted worst case rises to 5.16 $\log_{10}$, a margin of 1.27.

XMuLV clearance and step yield were robust to both parameters over the ranges studied. No model term reached significance for either response, and the two response-surface fits are reported as evidence of robustness and are not used predictively. Predicted XMuLV clearance remains at or above 5.99 $\log_{10}$ everywhere in the characterized region, against a required step contribution of 3.65 $\log_{10}$, and predicted step yield remains at or above 97.5%.

Both parameters are classified as well-controlled critical process parameters (WC-CPP), and the step carries no CPP and no key process parameter. Commercial-scale capability for the cumulative MVM clearance attribute is a one-sided Cpk of 1.51, with a margin of 1.78 $\log_{10}$ between the simulated mean and the requirement. No drug-substance attribute has a tighter capability, and the figure is cumulative over more than one step. Of the 10.03 $\log_{10}$ of MVM clearance claimed for the process, this step is credited with 5.32 $\log_{10}$ and anion exchange with 4.71 $\log_{10}$ (PCR-008).

No clearance of host cell protein, DNA, aggregate or leached Protein A is claimed for this step, and no product-quality attribute is formed here. The step does not meet the cumulative MVM requirement of 8.60 $\log_{10}$ on its own, and its clearance is credited only as one module of an orthogonal set. The campaign recorded 2 deviations at this step. Each was investigated, retained and impact-assessed, and none changed a parameter classification or the operating region (§13). The results of this step roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified by the platform train given in Table 1.1. The primary mechanism of action is antibody-dependent cellular cytotoxicity, so the glycan attributes that govern effector function are formed in the production bioreactor and are unaffected by the purification steps that follow. Viral safety, by contrast, is built across the purification train and not at any single step.

Table 1.1: The A-Mab drug-substance process train, with each step's principal role in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Small-virus retentive filtration is the last purification operation before formulation, and it works by passing the anion-exchange pool through a small-virus retentive membrane under constant pressure, with the filtrate collected as the load for ultrafiltration and diafiltration (Step 10). Retention is size based. Particles larger than the membrane retention rating are held back while the antibody monomer passes, so the mechanism does not depend on solution chemistry and is orthogonal to the mechanisms of the two other clearance steps. Low-pH inactivation (PCR-006) destroys the lipid envelope of enveloped viruses, and anion exchange (PCR-008) removes virus by charge-based binding in flow-through mode. A non-enveloped parvovirus such as MVM is not inactivated by low pH at all, which is why size-based removal carries the larger part of the MVM claim.

Two model viruses were used, in line with the practice of the case study (CMC Biotech Working Group 2009). MVM (minute virus of mice) is the smaller of the two and its diameter is closest to the retention rating of the membrane, so it is the worst-case challenge for a size-based step and the response that governs the operating region (§6). XMuLV (xenotropic murine leukaemia virus) is a large enveloped retrovirus model and is retained with a wide size margin.

1.2 Objectives

The characterization had five objectives. Three of them concern the measurements: (i) to quantify the relationship between the two characterized operating parameters and the log reduction achieved for MVM and XMuLV, (ii) to define the operating region over which the step meets its required contribution to the cumulative clearance claim, and (iii) to establish a proven acceptable range for each parameter against each governed attribute. The remaining two concern the use made of those measurements: (iv) to classify each parameter under the decision logic of SOP-4001, and (v) to justify the ranges that will be applied in Stage 2 process performance qualification and in the commercial control strategy for the step.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). Its record is organized so that the design, the execution, the analysis and the resulting ranges can be read as one traceable argument. The enhanced development approach of ICH Q8 and ICH Q11 provides the design-space framework, and ICH Q9 provides the risk methodology used in RA-001. Criticality is treated as a continuum in the sense of ICH Q9, so an attribute is not simply critical or non-critical but carries a level of criticality that reflects both impact and residual uncertainty.

Viral clearance is evaluated under ICH Q5A(R2) (International Council for Harmonisation 2023). Three consequences follow for this report. Clearance is demonstrated on a qualified small-scale spiking model, because virus cannot be introduced into the manufacturing facility. Clearance is claimed modularly, so each step is studied independently and the cumulative claim is the sum of the independent contributions. The conditions under which the reduction was demonstrated become part of the control strategy, so a parameter with no measurable effect on clearance is still held within the range in which the claim was established.

1.4 Related documents

This report is one of the unit-operation reports in the A-Mab characterization package. Its parents (PTP-001, RA-001 and PCMP-001), its paired plan (PCP-009) and the report it rolls up into (PCMR-001) are listed in Table 1.2.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-009	Process Characterization Plan (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Small-virus retentive filtration is a mature platform operation at Novacyte Biologics, and the study was designed to confirm and bound its behaviour rather than to discover it. The same membrane chemistry and the same mode of operation have been used for three prior humanized IgG1 products (X-Mab, Y-Mab and Z-Mab), and the mechanism of removal is identical across those products because retention depends on particle size and not on the antibody. Prior knowledge therefore transfers to A-Mab as a mechanistic expectation. It does not transfer as a numerical claim, and every log reduction credited to A-Mab comes from the studies reported here.

Platform experience sets two expectations that the study was built to test. The first is that the log reduction achieved for the smallest model virus declines as the cumulative volume filtered per unit membrane area increases. Protein and particulate matter deposit progressively on and within the membrane during a run, and the retention of particles whose size is close to the retention rating is the first to be affected. The second expectation is that operating pressure changes the flux and therefore the duration of the filtration, but does not change the pore size distribution that produces retention. Pressure was expected to be inactive on both clearance responses over the range studied.

2.2 Quality attributes in scope

The step governs the two viral-clearance attributes listed in Table 2.1. Neither is formed at this step. Both are cumulative attributes of the drug substance, accumulated across the purification train, and both are of very high criticality (VH) because a shortfall in viral clearance is a patient-safety issue with no downstream remedy. The step forms no product-quality attribute of its own, so the attributes that are formed upstream (the glycan attributes, the charge variants and aggregate) were not measured here.

Table 2.1: Viral-clearance quality attributes governed by the step, with the cumulative acceptance criteria for the drug substance and the criticality assigned under Tool #1.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

The acceptance criteria in Table 2.1 are cumulative and apply to the whole process. The criterion that applies to this step is the difference between the cumulative requirement and the clearance credited to the other contributing steps, and it is derived in §7. Step yield was measured as a process-performance response. It carries no drug-substance specification and is reported here as evidence that the operating region does not cost product.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every operating input of the step on its potential impact on a CQA and on its potential to interact with other inputs, and it assigned a study type to each. Two parameters scored high enough for multivariate study: filtration volume, expressed as the cumulative load per unit membrane area (L/m^2), and filtration pressure (psi). The remaining inputs are held to platform specification and were verified rather than characterized. Membrane lot and effective filtration area are controlled by pre-use flux verification and by the integrity test described in SOP-2012, and the composition of the load is fixed by the anion-exchange pool specification agreed in PCR-008.

The ranges taken into the study are wider than the ranges the process will run in, which is the standard basis for a proven acceptable range (CMC Biotech Working Group 2009). Filtration volume was characterized over 50–140 L/m^2, which is 1.64 times the width of its normal operating range of 50–105 L/m^2. The upper edge was placed well above the routine limit deliberately, because the expected decline in MVM retention with load can only be bounded by driving the step past the load it will see. Filtration pressure was characterized over 8–30 psi, 2.20 times the width of its normal operating range of 10–20 psi. The pressure range brackets the set-point on both sides and covers the excursions that a pump ramp can plausibly produce.

3 Materials and methods

3.1 Scale-down model and its qualification

All data reported here were generated on a scale-down model of the commercial filtration step, qualified under SOP-1001. The model uses the same membrane, the same feed material and the same mode of operation as the commercial step, at a reduced membrane area. Scale is held by the two quantities that determine performance for a size-based step: the cumulative volume filtered per unit membrane area (L/m^2), and the applied pressure (psi). Both are scale-independent, so a run at a given load and pressure on the small-scale device is equivalent to a commercial run at the same load and pressure.

Qualification compared unspiked small-scale runs at the set-point with at-scale performance on the same measures the commercial step reports (flux decay across the run, step yield and post-use integrity), and no difference of practical consequence was found on any of them. The qualification record is held under SOP-1001 and summarized in PCP-009. The scale-down model is therefore suitable for defining operating ranges and for supporting a clearance claim.

Spiking cannot be performed at commercial scale, so every log reduction reported here comes from the small-scale model. This is the arrangement ICH Q5A(R2) expects, and it carries a condition (International Council for Harmonisation 2023). The clearance claim is only as good as the representativeness of the model, so the qualification above is part of the claim and not a preliminary to it. The limitation this leaves is stated in §11.

3.2 Operation

Each run followed SOP-2012. The membrane was flushed and integrity tested before use, the anion-exchange pool was spiked with the model virus and pre-filtered, and the spiked load was then filtered at constant pressure until the target volumetric load had been delivered. Filtrate was collected as a single pool. The membrane was integrity tested again after use, and a run that failed the post-use test would have been excluded from the analysis, since a compromised device tells nothing about the retention of an intact one. No run failed.

Load material for the study was the anion-exchange pool produced under the conditions established in PCR-008. Using the real intermediate matters for a fouling-driven response, because the deposition that reduces retention at high load is caused by the protein and particulate content of the actual feed. Virus spike level, spike volume fraction and pre-filtration were held constant across all runs so that the measured reduction reflects the membrane and the operating conditions only.

3.3 Analytical methods

The controlled procedures and validated methods used by the step are listed in Table 3.1. Infectivity titres in the load and in the filtrate were determined for both model viruses, and the log reduction factor for each run is the difference between the total infectivity loaded and the total infectivity recovered in the filtrate. MVM titre was determined under AMV-3018 by a fifty per cent tissue culture infectious dose assay with a quantitative polymerase chain reaction confirmation, and XMuLV titre under AMV-3017 by a fifty per cent tissue culture infectious dose assay. Step yield was determined by mass balance from protein concentration and volume.

Table 3.1: Controlled procedures and validated analytical methods applied to the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

Both infectivity assays are validated, and their reported precision is the dominant component of the run-to-run scatter in the clearance responses. Method performance is summarized in Appendix D. The MVM assay (AMV-3018) has a reported precision of 0.32 $\log_{10}$ and contributes about 30.0% of the observed variance in the MVM response. The corresponding figures for the XMuLV assay (AMV-3017) are 0.3 $\log_{10}$ and 28.0%. Effects of a size comparable to the assay precision are therefore treated as unresolved, and are reported as such in §5.

3.4 Sampling plan

Each run was sampled at three points: the unspiked pool before spiking, the spiked load after pre-filtration, and the filtrate pool at the end of the run. The spiked load and the filtrate pool provide the two titres from which the log reduction is calculated, and the unspiked pool confirms the absence of adventitious signal in the assay. Protein concentration was measured in the load and in the filtrate for the yield calculation.

Replication was placed at the centre of the design. The screening design carried 3 centre-point runs and the response-surface design carried 4, all at the design centre of both factors. These replicates provide the pure-error estimate against which lack of fit is tested, and they are the only source of replication in either design.

3.5 Statistical methods

Both designs were analysed in coded factors, with the centre of the characterization range at 0 and its edges at −1 and +1, so that estimated effects are directly comparable between parameters of different units. The screening design was fitted by ordinary least squares with both main effects and their interaction, and an effect is reported as twice the coded coefficient, which is the change in response across the full characterized range of the parameter. The response-surface design was fitted with the full quadratic model in both factors. Significance was assessed at $\alpha = 0.05$ throughout, following SOP-1002.

Model adequacy was judged on four criteria taken together: the overall model F test, the coefficient of determination and its adjusted form, the predicted coefficient of determination computed from the prediction sum of squares, and an analysis of variance partition of the residual into lack of fit and pure error. A model whose lack of fit is significant against pure error is not used to define a region. A response for which no term reaches significance is reported as robust over the ranges studied, and is not given a predictive model.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with every parameter of the process drawn independently across its normal operating range and the fitted models used to propagate those settings to the drug-substance attributes. Capability against a one-sided lower requirement is reported as Cpk. The simulation reflects parameter variation and model uncertainty, and it does not reflect the batch-to-batch variation of a commercial facility, which is confirmed in Stage 2.

4 Study design

4.1 Factors, ranges and the knowledge space

The design covers two parameters, and that is the correct size for this step. Only two of the step's operating inputs can plausibly affect a size-based retention mechanism, and RA-001 identified both. The parameters, their set-points, their normal operating ranges, their characterized ranges and their final classification are given in Table 4.1, and the coded letters used in the effect and coefficient tables are given in Table 4.2.

Table 4.1: Characterized parameters of the step, with set-point, normal operating range, characterization range, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Filtration volume (load)	L/m2	90	50–105	50–140	WC-CPP	multivariate
Filtration pressure	psi	13	10–20	8–30	WC-CPP	multivariate

Table 4.2: Factor legend for the coded designs.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

The characterized ranges define the knowledge space of the step. The design space is a subset of that knowledge space in which the governed attributes stay within their acceptance criteria, and the normal operating ranges sit inside the design space (CMC Biotech Working Group 2009). The load range extends above the routine limit by design, so that the knowledge space contains the region in which MVM retention is expected to degrade. The pressure range extends both above and below the routine limits, because pressure excursions in either direction are physically plausible on a constant-pressure filtration.

4.2 Screening design

The screening study was a two-level full factorial in both parameters with 3 centre points, 7 runs in total. Runs were randomized. With two factors a full factorial estimates both main effects and their interaction with no aliasing at all, so this design has none of the resolution limitations that constrain a fractional screen. Its purpose was to establish which parameters are active, and to do so on an independent set of runs before the response-surface study was committed. The design matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same two parameters, 12 runs with 4 centre points. The axial points sit at the edges of the characterized ranges, so the design adds curvature without extending the knowledge space beyond the ranges already justified in §2.3. The fitted model carries 5 terms besides the intercept, which leaves 6 residual degrees of freedom, of which 3 are pure error.

The division of labour between the two designs needs to be stated plainly, because it is not the usual one. In a step with four or five factors the screening design is near-saturated and can only identify which factors matter, while the response-surface model is the predictive model and the basis of the design space. Here the screening design is not saturated, and it estimates the same parameters as the response-surface model does, minus the two quadratic terms. The response-surface model is still the predictive model and the basis of the design space, for two reasons. It uses the axial runs to test whether the response is linear in load across the full range, and it carries a larger replicated centre and therefore a better estimate of pure error. The screening study is used here as an independent confirmation of the sign and approximate size of the load effect, and its estimates are reported for that purpose only.

4.4 Univariate assessment

No parameter of this step was assessed univariately, because both characterized parameters went into the multivariate design. Three further inputs were verified against platform specification without a designed study. Membrane lot permeability is verified before use by a flush and flux measurement, effective filtration area is fixed by the device, and the composition of the load is controlled by the anion-exchange pool specification. These are verified and not characterized because none of them is an adjustable operating parameter of the step. A finding against one of them is a material or equipment finding and is handled as a deviation, which is what happened in DEV-009-02 (§13.2).

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility at the design centre is good for all three responses, and it is poorest for the MVM log reduction factor. The centre-point means, standard deviations and coefficients of variation are given in Table 5.1 for the screening design and Table 5.2 for the response-surface design.

Table 5.1: Centre-point performance in the screening design: mean, standard deviation and coefficient of variation per response.

Response	n	Mean	SD	%CV
MVM LRF (log_{10})	3	5.44	0.488	8.99
XMuLV LRF (log_{10})	3	6.3	0.147	2.34
Step yield	3	0.986	0.00767	0.777

Table 5.2: Centre-point performance in the response-surface design, which provides the pure-error term for the lack-of-fit tests.

Response	n	Mean	SD	%CV
MVM LRF (log_{10})	4	5.54	0.116	2.1
XMuLV LRF (log_{10})	4	6.13	0.176	2.87
Step yield	4	0.983	0.00839	0.854

The MVM response is the least precise of the three in both designs. In the screening design its centre-point coefficient of variation was 8.99 per cent, corresponding to a standard deviation of 0.49 log_{10}. That is of the same order as the reported precision of the MVM infectivity assay (0.32 log_{10}, Appendix D), so most of the scatter at the centre of the screening design is analytical. In the response-surface design, with one more replicate, the corresponding coefficient of variation was 2.10 per cent and the standard deviation 0.12 log_{10}. Two things follow. The pure-error term used in the lack-of-fit tests below is estimated from 4 replicates and should be read as a lower bound on the true run-to-run variation. Any effect on MVM clearance smaller than about the assay precision is beyond the resolution of this study and is not claimed.

XMuLV and step yield reproduce more tightly. The XMuLV centre-point coefficient of variation was 2.87 per cent and the step-yield value 0.85 per cent in the response-surface design. For step yield in particular, the centre-point scatter is small enough that a factor effect of any practical consequence would have been detected.

5.2 Screening: factor effects

Volumetric load reduces MVM clearance, and it is the only parameter whose effect was resolved on any response. The screening model fits are summarized in Table 5.3 and the MVM effect estimates in Table 5.4; the corresponding tables for XMuLV and step yield are in Appendix C.

Table 5.3: Screening model summary for each response.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	7	0.848	0.695	0.769	5.56	0.0962	0.4
XMuLV LRF ($\log_{10}$)	7	0.714	0.428	0.364	2.5	0.236	0.122
Step yield	7	0.709	0.418	-2.23	2.43	0.242	0.0073

Table 5.4: Screening effect estimates for the MVM log reduction factor. An effect is the change in response across the full characterized range of the parameter.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-1.61	-0.807	0.2	-4.04	0.0273	*
B	-0.246	-0.123	0.2	-0.617	0.581	
AB	0.0192	0.00962	0.2	0.0482	0.965	

Load carried an effect of -1.61 $\log_{10}$ across its characterized range, with $p = 0.027$. Pressure carried -0.25 $\log_{10}$ and the interaction 0.02 $\log_{10}$, neither of which approaches significance. The overall F test for the MVM screening model does not reach significance either ($p = 0.096$), which is the expected outcome when two of a model's three terms are inactive and only 3 residual degrees of freedom remain. The load term is significant on its own, and that is the finding the screening study was run to produce.

Neither XMuLV clearance nor step yield gave a significant term ($p = 0.236$ and $p = 0.242$ for the respective models). The largest XMuLV effect was that of pressure, at 0.26 $\log_{10}$ across the characterized range, which is not distinguishable from the assay precision. These are null results, and they were carried forward as findings. Both parameters were retained in the response-surface design so that the robustness of XMuLV clearance and yield would be tested again on an independent set of runs and over the axial points.

5.3 Response-surface models

Only the MVM model is adequate for prediction. The response-surface model summaries are given in Table 5.5.

Table 5.5: Response-surface model summary for each response.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	12	0.947	0.903	0.527	21.4	0.000921	0.211
XMuLV LRF ($\log_{10}$)	12	0.506	0.0949	-1.14	1.23	0.398	0.158
Step yield	12	0.26	-0.356	-1.56	0.422	0.819	0.00685

The MVM model explains 94.7% of the variation in the response, with an adjusted value of 90.3% and an overall F test of $p < 0.001$. Its root mean square error is 0.21 $\log_{10}$, which is close to the centre-point standard deviation reported above. The predicted coefficient of determination is 52.7%, well below the adjusted value. That gap is a real limitation and it is what a design of 12 runs with 5 model terms produces under leave-one-out cross-validation. The model is used to predict mean levels over the ranges studied, and it is not used to extrapolate beyond them.

The XMuLV and step-yield models are not adequate and are not used predictively. The XMuLV model returns $p = 0.398$ with an adjusted coefficient of determination of 9.5%, and the step-yield model $p = 0.819$. Neither result is a defect in the study. A response that no parameter moves cannot produce a significant model, and the correct reading is that both responses are robust to load and pressure over the ranges studied. The evidence for that reading is given at the end of this section.

Table 5.6: Full quadratic coefficient table for the MVM log reduction factor, in coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.52	0.0962	57.4	1.87e-09	***
A	-0.866	0.086	-10.1	5.57e-05	***
B	-0.0341	0.086	-0.396	0.706	
AB	0.028	0.105	0.266	0.799	
A^2	0.136	0.129	1.05	0.333	
B^2	-0.301	0.129	-2.34	0.0582	

Load dominates the MVM model. Its coded coefficient is -0.87 $\log_{10}$ per half-range with $p < 0.001$ (Table 5.6), and no other term reaches significance. The quadratic term in pressure is the largest of the remainder at -0.30 with $p = 0.058$, and it is not claimed. Translated into natural units, the model predicts 6.47 $\log_{10}$ at the bottom of the characterized load range, 5.55 at the set-point of 90 L/m², 5.26 at the upper limit of the normal operating range, and 4.71 at 140 L/m². The decline is close to linear, which is why the quadratic term in load is not significant.

Table 5.7: Analysis of variance for the MVM response-surface model, partitioning the residual into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	4.75	5	0.951	21.4	0.000921
Residual	0.266	6	0.0444	nan	nan
Lack of fit	0.226	3	0.0752	5.55	0.0965
Pure error	0.0406	3	0.0135	nan	nan

Lack of fit for the MVM model does not reach significance, but the margin is not comfortable. The partition is given in Table 5.7, and the lack-of-fit test returns F = 5.55 with p = 0.096 against 3 degrees of freedom of pure error. The test is passed at $\alpha = 0.05$ and the model is accepted, with the reservation that a pure-error estimate from 4 replicates is itself imprecise and that a small pure error inflates the lack-of-fit ratio. The corresponding tests for XMuLV (p = 0.655) and step yield (p = 0.805) are far from significance, as they should be for responses with no structure to fit.

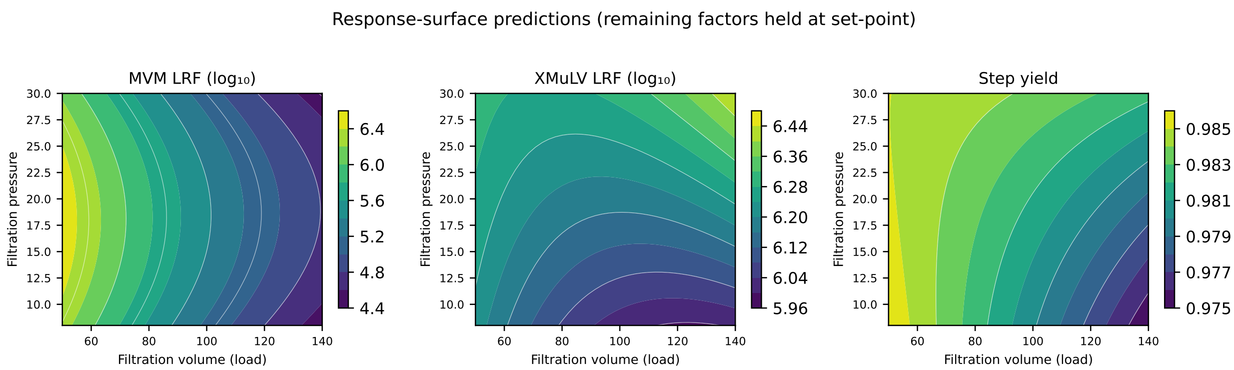

Figure 5.1: Response-surface predictions over filtration volume and filtration pressure for all three measured responses. The MVM surface falls steadily with load and is almost flat in pressure; the XMuLV and step-yield surfaces are flat in both parameters.

The contour panel in Figure 5.1 makes the shape of the three surfaces plain. The MVM surface is a set of near-vertical bands, so the response is governed by load and almost independent of pressure. The XMuLV and step-yield surfaces are flat over the whole rectangle, and their apparent gradients are smaller than the replicate scatter reported in §5.1.

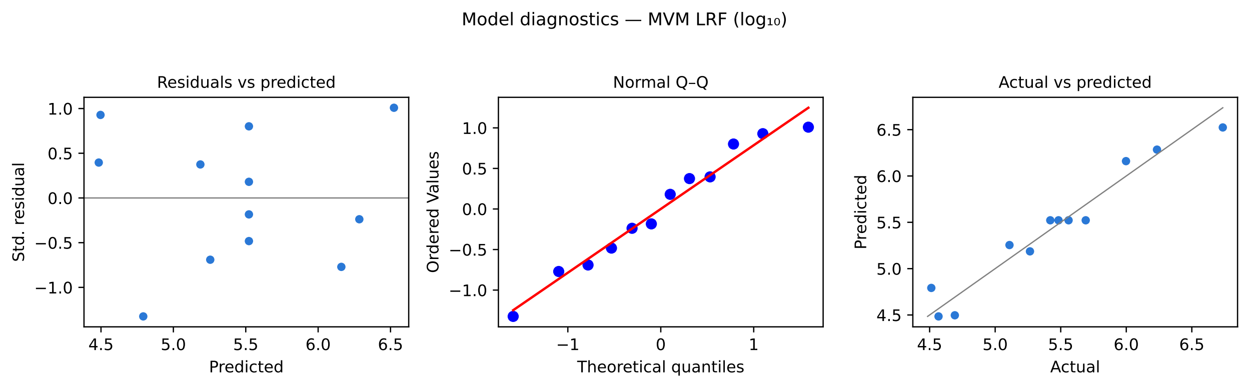

Figure 5.2: Model diagnostics for the MVM log reduction factor: standardized residuals against predicted values, a normal quantile plot of the residuals, and actual against predicted values.

Diagnostics for the MVM model are shown in Figure 5.2. The residuals show no trend against the predicted value and no funnel, the normal quantile plot is close to a straight line, and the actual values fall on the identity line across the whole range of prediction. There is no evidence of a missing term or of a variance that changes with the level of the response.

5.4 Mechanistic interpretation

The shape of the MVM surface follows from how a size-based membrane retains a particle whose diameter is close to its retention rating. Retention is not a step function of size. The membrane presents a distribution of flow paths, and MVM is small enough that it is held by the tighter part of that distribution and passed by the loosest part. As filtration proceeds, protein and particulate matter from the load deposit within the membrane and progressively change that distribution, and the fraction of the load that finds a path through increases with the volume that has already been filtered. The result is a log reduction that decreases steadily with cumulative volumetric load, which is what Table 5.6 and Figure 5.1 show, and which is consistent with the behaviour reported for this class of filter in the case study (CMC Biotech Working Group 2009).

XMuLV is retained by a wide margin because its diameter is far above the retention rating, so the change in the flow-path distribution that costs MVM retention leaves XMuLV unaffected. This is the expected result and the study confirms it. The absence of a load effect on XMuLV also argues against gross membrane damage as the mechanism behind the MVM trend, since damage severe enough to pass a parvovirus at higher rates would be expected to pass a retrovirus as well.

Pressure has no resolved effect on either clearance response over 8–30 psi. That is consistent with a size-based mechanism, in which pressure sets the flux and hence the duration of the run but does not alter the pore size distribution. It is also consistent with the step-yield result, since the antibody monomer passes freely at every pressure studied. The practical consequence is that the operating region of this step is effectively one-dimensional. It is bounded by load, and pressure is constrained by the clearance claim and not by a measured effect. That is a narrower structure than the multivariate regions reported for the chromatographic steps, and it is stated here so that the design space in §6 is not read as more complex than the data support.

6 Design space

The design space of the step is the region of the characterized ranges over which the predicted MVM log reduction meets the required step contribution, and that region is the whole characterized rectangle. Across a grid spanning 50–140 L/m^2 and 8–30 psi, 100.0% of the predicted MVM values meet the step requirement of 3.89 $\log_{10}$, and 100.0% of the predicted XMuLV values meet theirs. The principal plane of the region is load against pressure, which is the only plane there is.

The worst case within the characterized region is the highest load, at 140 L/m^2 and 30 psi. The MVM log reduction predicted there is 4.48 $\log_{10}$, a margin of 0.59 over the step requirement. The worst case within the normal operating ranges is 105 L/m^2 at 10 psi, where the predicted value is 5.16 $\log_{10}$ and the margin 1.27. Both worst cases are at the top of the load range, as the mechanism in §5.4 predicts.

The clearance credited to the step in the cumulative claim is 5.32 $\log_{10}$ for MVM and 5.82 $\log_{10}$ for XMuLV. The response-surface model predicts 5.55 $\log_{10}$ for MVM at the set-point, so the credited value sits 0.23 $\log_{10}$ below the prediction at the condition the process will run at. The credit is conservative with respect to the characterization data, which is the appropriate direction for a safety claim.

Three ranges are nested at this step, and they should not be confused with one another. The normal operating ranges of 50–105 L/m^2 and 10–20 psi are the ranges the batch record will hold. The characterized ranges of 50–140 L/m^2 and 8–30 psi are the knowledge space. The design space is the region within the knowledge space that meets the step requirement, and here it covers the whole knowledge space. Movement within the design space is not a change in the sense of ICH Q8(R2), and movement outside it requires change control under the pharmaceutical quality system of ICH Q10.

Three bounds apply to this claim. It holds over the characterized ranges only, and no statement is made about loads above 140 L/m^2 or pressures outside 8–30 psi. It rests on a mean-level model, so the assurance statement that accounts for prediction uncertainty is the one in §7 and not this section. It rests on the scale-down model qualified in §3.1, and the edges of the region have not been confirmed at commercial scale.

7 Proven acceptable ranges

The design space above is a multivariate region. It is complemented here by proven acceptable ranges, which are the ranges of each parameter individually over which the governed attribute stays within acceptance. A viral-clearance attribute carries no drug-substance specification that a single step can be measured against, so the acceptance basis is derived below before any range is claimed.

The acceptance criteria in Table 2.1 are cumulative requirements on the whole process. The criterion that applies to this step is the cumulative requirement minus the clearance credited to every other contributing step, which is the modular arithmetic of ICH Q5A(R2) (International Council for Harmonisation 2023). For MVM the cumulative requirement is 8.60 $\log_{10}$ and anion exchange is credited with 4.71 $\log_{10}$ (PCR-008), which leaves 3.89 $\log_{10}$ for this step. Low-pH inactivation is credited with no MVM clearance at all, because a non-enveloped parvovirus is not inactivated by low pH. For XMuLV the cumulative requirement is 16.70 $\log_{10}$, low-pH inactivation is credited with 7.10 $\log_{10}$ (PCR-006) and anion exchange with 5.95 $\log_{10}$, which leaves 3.65 $\log_{10}$ here. These two floors are the acceptance criteria used throughout this report.

Table 7.1: Proven acceptable ranges for each governed attribute and each parameter, at the set-point and with the other parameter varying within its normal operating range.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
MVM LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	50–140	50–140
MVM LRF ($\log_{10}$)	Filtration pressure	8–30	psi	8–30	8–30
XMuLV LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	50–140	50–140
XMuLV LRF ($\log_{10}$)	Filtration pressure	8–30	psi	8–30	8–30

Every proven acceptable range in Table 7.1 spans the full characterization range of its parameter, under both analyses. The first analysis holds the other parameter at its set-point and scans the parameter of interest across its characterized range. The second repeats the scan with the other parameter drawn at random from its normal operating range, propagating that variation through the fitted model by Monte-Carlo simulation of 2,000 draws at each of 81 grid points. The two analyses agree for all four combinations (Table 7.1), which means that no parameter of this step has an edge of failure inside its characterized range, and that the result is not sensitive to how the other parameter moves within its routine range.

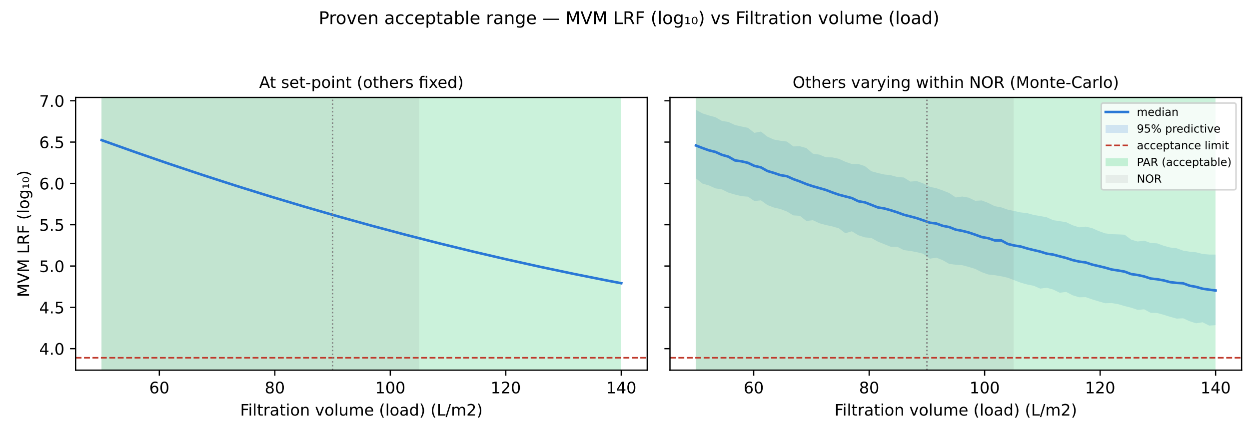

Figure 7.1: Proven acceptable range for the MVM log reduction factor against filtration volume. Left panel: filtration pressure held at its set-point. Right panel: filtration pressure varying within its normal operating range, with the median and the 95 per cent predictive band. The acceptance limit is the required step contribution and the acceptable region is shaded green.

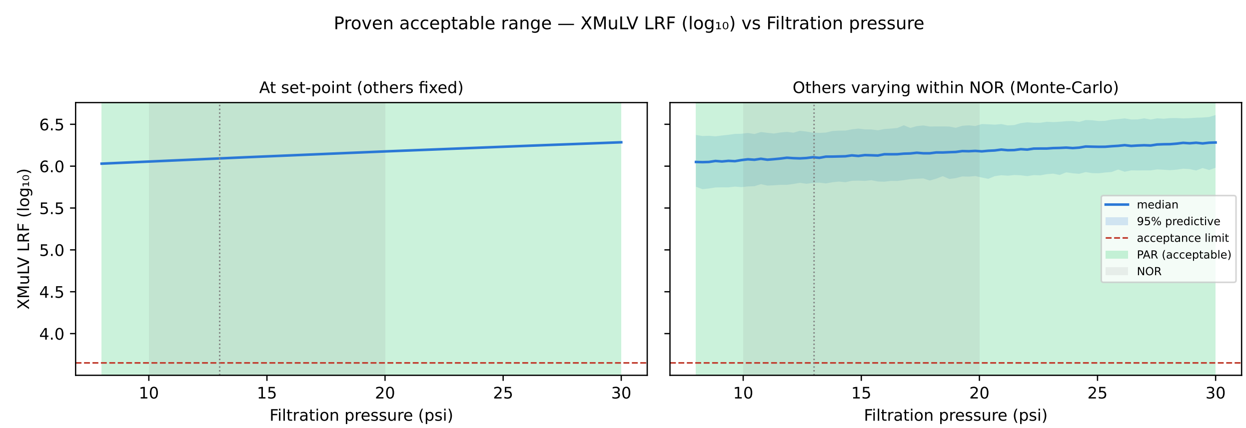

Figure 7.2: Proven acceptable range for the XMuLV log reduction factor against filtration pressure, in the same two-panel form. The response is flat and the acceptance limit is far below the predicted values across the whole characterized range.

The adverse trend is drawn against its acceptance limit in Figure 7.1. The predicted MVM log reduction falls steadily from left to right, the acceptance limit runs well below the curve, and the green region covers the whole axis. The right panel adds the information that matters for assurance. With pressure varying across its normal operating range, the lower edge of the 95 per cent predictive band is 4.83 $\log_{10}$ at the upper limit of the normal operating range for load, and 4.28 $\log_{10}$ at 140 L/m². The latter is still 0.39 $\log_{10}$ above the step requirement, and the simulated assurance of meeting the requirement does not fall below 100.0% anywhere on the axis. Figure 7.2 shows the corresponding picture for XMuLV, where the response is flat and the margin to the limit is large throughout.

Two consequences follow for how the ranges are used. Because every proven acceptable range covers the full characterized range, no univariate range binds the operating region; the binding constraint on the step is the worst-case corner of the multivariate region identified in §6, and in practice the load limit written into the batch record. Because the proven acceptable ranges were established against a step requirement that is itself derived from the clearance credited elsewhere, they are only valid while that credit holds. A change to the clearance claimed for anion exchange would change the floor for this step and would require the ranges here to be re-derived.

Both conclusions are bounded by the characterization ranges of 50–140 L/m^2 and 8–30 psi, and by the fitted response-surface model. Neither is a statement about conditions outside those ranges, and neither is a statement about a membrane type other than the one used here.

8 Process capability and robustness

The step supports both cumulative viral-clearance attributes with adequate capability, and the MVM attribute is the tightest capability of the drug substance. Capability was estimated by Monte-Carlo simulation of 2,000 batches with every process parameter drawn across its normal operating range. Results for the two attributes this step governs are given in Table 8.1.

Table 8.1: Commercial-scale capability for the two viral-clearance attributes, from a Monte-Carlo simulation of the whole process with all parameters inside their normal operating ranges.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

Cumulative MVM clearance has a mean of 10.38 $\log_{10}$ against a requirement of 8.60 $\log_{10}$, a margin of 1.78 $\log_{10}$, and a one-sided Cpk of 1.51. The lowest simulated batch reached 9.08 $\log_{10}$, which is still above the requirement. Cumulative XMuLV clearance is less constrained, with a mean of 19.51 $\log_{10}$, a margin of 2.81 and a Cpk of 1.53. Both exceed the value of 1.33 that is conventionally taken as adequate for a critical characteristic. The minimum Cpk recorded across all drug-substance attributes is 1.51, and it belongs to the MVM attribute.

That minimum belongs to the process and not to this step. The MVM figure in Table 8.1 is cumulative over the two steps credited with MVM clearance, of which this step contributes 5.32 $\log_{10}$ and anion exchange 4.71 $\log_{10}$ (PCR-008). This step therefore carries about 53.0% of the claim, and it is the larger of the two contributions. The XMuLV figure is cumulative over three steps, of which low-pH inactivation contributes the most (PCR-006) and this step 5.82 $\log_{10}$, or about 30.8% of the total.

Robustness of the step itself is separate from capability, and the evidence for it is the three null results in §5. Neither parameter affects XMuLV clearance or step yield over 50–140 L/m² and 8–30 psi, and only load affects MVM clearance. Predicted step yield does not fall below 97.5% anywhere in the characterized region, against 98.2% at the set-point, so the operating region costs no measurable product.

These capability figures are bounded in two ways. They are computed with every parameter inside its normal operating range, so they describe routine operation and not operation at the edges of the design space, and they are computed from qualified scale-down models rather than from manufacturing data, so they will be confirmed against commercial-scale results during Stage 2 process performance qualification.

9 Parameter classification

Both characterized parameters are classified as well-controlled critical process parameters under SOP-4001. The classification and the reason for each are as follows.

Filtration volume, expressed as load per unit membrane area, is a WC-CPP. It has a demonstrated effect on MVM clearance, which is a very high criticality attribute, so it is quality-linked and cannot be a general process parameter. It is not a CPP, because the volume delivered per unit membrane area is set by the batch record and measured directly during the run, so the risk of leaving the design space is low. The parameter is easy to control and the consequence of an excursion is graded and not abrupt, as Figure 7.1 shows directly.

Filtration pressure is also a WC-CPP, and the reason is different. No effect of pressure was resolved on any of the three responses over 8–30 psi, so on the measured evidence alone it would be a general process parameter. It is classified WC-CPP because the viral clearance claimed for this step was demonstrated under the conditions studied, and ICH Q5A(R2) ties the claim to those conditions (International Council for Harmonisation 2023). Holding pressure within 10–20 psi keeps the step inside the range in which the claim was established. Pressure is straightforward to control on a constant-pressure filtration, so the risk of leaving the range is low.

The null results were retained in the knowledge space. The absence of a pressure effect on MVM and XMuLV clearance, and the absence of any effect on step yield, are evidence that the step is robust to both parameters over the ranges studied, and they are the basis for the wide characterized ranges recorded in Table 4.1. They are also the reason no additional in-process control is proposed for pressure beyond the normal operating range.

The step carries no CPP and no key process parameter. Across the whole drug-substance process, 20 parameters are classified WC-CPP and 1 is classified CPP, so both parameters of this step fall into the larger class, and the one CPP of the process belongs to low-pH inactivation (PCR-006).

10 Contribution to the control strategy

The step contributes to the control strategy in four ways, and it is the principal mechanistic barrier to parvovirus in the process. Its contributions are the following.

Volumetric load is held at or below 105 L/m² by the batch record, which specifies the membrane area to be installed and the volume to be filtered. This is the control that preserves the MVM claim, and it is the one control at this step that follows from a measured effect.

Filtration pressure is held within 10–20 psi and is monitored continuously during the run. The control exists to keep the step within the conditions under which the clearance was demonstrated, not because a pressure effect was found.

Membrane integrity is verified before and after every run under SOP-2012. A post-use integrity failure invalidates the clearance claimed for that batch, and this is the control that detects the failure mode a designed experiment cannot cover, which is a defective or damaged device.

Membrane lot permeability is verified before use by a flush and flux measurement. The control was already in place and its value was demonstrated during this campaign, when it detected the lot described in §13.2.

Viral clearance is not a batch release test. It is established by the validation studies reported here and in PCR-006 and PCR-008, so the assurance of viral safety for a commercial batch rests entirely on holding the validated parameter ranges and on the integrity tests above. This is why a parameter with no measured effect is still controlled here.

Two limits on this step's contribution should be read alongside the list. The step is not credited with clearance of host cell protein, DNA, aggregate or leached Protein A, and no such claim is made anywhere in this report; those attributes are controlled by Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008). The step also does not deliver the cumulative MVM requirement by itself, since its credited 5.32 $\log_{10}$ is below the cumulative requirement of 8.60 $\log_{10}$. The modular claim is consolidated in PCMR-001, which is where the cumulative arithmetic and the orthogonality argument are presented for the process as a whole.

11 Discussion

The step is well understood, and the understanding is simple because the mechanism is simple. One parameter governs the one response that varies, and the direction of the effect was predicted by prior knowledge before the study was run. Volumetric load reduces MVM clearance by progressively changing the flow-path distribution of the membrane, and the study quantified that relationship over a load range 1.64 times the width of the routine range. The result is a step whose principal risk is known, bounded and controlled by a single entry in the batch record.

The adverse finding deserves to be read in its own terms before its mitigation. MVM clearance at the top of the characterized load range is 1.76 $\log_{10}$ below its value at the bottom, and that is a large change for an attribute whose cumulative capability is the tightest of the drug substance. The mitigation is that the step retains a margin of 0.59 $\log_{10}$ over its required contribution even at the worst corner studied, and a margin of 1.27 $\log_{10}$ at the worst corner of the normal operating ranges. The residual position is that the load limit is a real constraint on the process and must be treated as one, and that operating above the characterized range has not been studied and is not supported.

Confidence in the scale-down model is high for this step, and higher than it would be for a chromatographic step. Scale enters a size-based filtration through two quantities only, which are the volume filtered per unit membrane area and the applied pressure, and both are matched exactly between the small-scale device and the commercial installation. The qualification described in §3.1 compared the unspiked performance of the two, and no difference of practical consequence was found.

Five limitations qualify the conclusions of this report. The response-surface model is a mean-level model whose predicted coefficient of determination (52.7%) is well below its adjusted value, so predictions near the edge of the region carry more uncertainty than the fit alone suggests; the assurance statement in §7 is the one that accounts for this. The pure-error term rests on 4 centre-point replicates, and the lack-of-fit test for the MVM model returns $p = 0.096$, which passes but not by a wide margin. All clearance data come from a small-scale spiking model, which ICH Q5A(R2) requires but which means the claim is conditional on the qualification of that model (International Council for Harmonisation 2023). The study used one membrane type and a limited number of membrane lots, so lot-to-lot permeability variation is managed by the pre-use verification described in §10 and not by the design space. The edges of the region have not been confirmed at commercial scale, and confirmation of the routine ranges is a Stage 2 activity.

Each of these is managed forward. The load limit and the pressure range enter the batch record and the PPQ protocol. The pre-use flux verification and the post-use integrity test remain in the control strategy for the life of the product. The clearance claim is revisited if the membrane, the load composition or the credited clearance of anion exchange changes.

No parameter of this step required the most stringent classification. Neither parameter has a narrow range, a steep response near a limit, or a control capability that would put the design space at risk, so neither is a CPP. The single CPP of the drug-substance process remains the inactivation pH of the low-pH step (PCR-006).

12 Conclusions

The small-virus retentive filtration step is well characterized over 50–140 L/m² of volumetric load and 8–30 psi of filtration pressure, and it meets its required contribution to the viral-clearance claim across that whole region. MVM clearance falls with volumetric load, by 1.76 $\log_{10}$ across the characterized range, and the worst corner of the region still delivers 4.48 $\log_{10}$ against a requirement of 3.89. XMuLV clearance and step yield are robust to both parameters over the same ranges.

The operating region proposed for Stage 2 is 50–105 L/m² of load and 10–20 psi of pressure. Both parameters are classified WC-CPP, and the step carries no CPP and no key process parameter. Commercial-scale capability for the cumulative MVM attribute is a Cpk of 1.51 with a margin of 1.78 $\log_{10}$ to the requirement, which is the tightest capability of the drug substance; the cumulative XMuLV attribute gives 1.53.

Three bounds attach to these conclusions. They hold over the characterized ranges and not beyond them. They rest on a qualified scale-down spiking model whose edges have not been confirmed at commercial scale. They rest on a modular clearance claim in which this step is credited with 5.32 $\log_{10}$ of MVM reduction and cannot meet the cumulative requirement alone. All 2 recorded deviations were investigated and retained, and none altered a classification or the operating region. The step's contribution is consolidated in PCMR-001.

13 Deviations from the plan

The study recorded 2 deviations from PCP-009 during execution. Each was investigated, assigned a root cause and impact-assessed against the study conclusions, and each was dispositioned as retained. None altered a parameter classification, the fitted models or the operating region. The register is given in Table 13.1 and each event is described below.

Table 13.1: Register of deviations recorded during execution of PCP-009.

Deviation	Summary	Detected during	Disposition
DEV-009-01	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	retained
DEV-009-02	Filter membrane lot flux below vendor-typical value	pre-use flux verification	retained

The two events are of different kinds and were investigated differently. DEV-009-01 is an excursion of a characterized parameter beyond its normal operating range during a study run, so its assessment is a question about the design. DEV-009-02 is a material finding on an input that the study did not characterize, so its assessment is a question about the representativeness of the runs it touched.

13.1 DEV-009-01 — filtration pressure excursion above the normal operating range

Filtration pressure reached 22 psi during a study run, which is 2 psi above the upper limit of the normal operating range of 10–20 psi. The excursion was transient and was detected by the in-process pressure monitoring required by SOP-2012.

The investigation attributed the excursion to overshoot on the feed pump ramp at the start of the run. The control loop approached its pressure set-point from above instead of asymptotically, and the peak occurred in the first minutes of the run, before any appreciable load had accumulated on the membrane. No equipment fault was found and the run completed normally, with a passing post-use integrity test.

The impact assessment turns on two facts. The peak pressure of 22 psi lies inside the characterized range of 8–30 psi, at a coded level of 0.27, so the condition experienced during the excursion is one the design space covers. Pressure has no resolved effect on any of the three responses (§5.2 and §5.3), and evaluating the fitted MVM model at the peak pressure changes the predicted log reduction by 0.04 $\log_{10}$ relative to the set-point, which is well below the precision of the MVM assay. The run was retained in the analysis. The corrective action was a revision of the pump ramp profile to approach the pressure set-point without overshoot, which is carried into the commercial batch record.

13.2 DEV-009-02 — membrane lot flux below the vendor-typical value

Membrane lot LOT-VF-9331 gave a water flux 12 per cent below the vendor-typical value at the pre-use flux verification. The finding was made before any product contacted the device, by the verification step described in §10.

The investigation attributed the deficit to lot-to-lot permeability variation within the vendor's release specification. The lot met its release criteria and its integrity test, and the deficit was in permeability and not in retention. These are separate membrane properties. Permeability is set by the number and the size of the flow paths available and determines how fast the load passes, whereas retention is set by the tightest part of the same distribution and determines what is held back. A lot with lower permeability filters more slowly at a given pressure, and it has no reason to retain less.

The affected runs were retained. Three lines of evidence support that disposition. The runs were executed at the same volumetric load per unit membrane area and the same pressure as the rest of the design, and those are the two quantities the scale-down model is built on (§3.1). The MVM titres for those runs were determined under AMV-3018 (Appendix D), and the residual diagnostics for the fitted MVM model show no outlying observation (Figure 5.2). The post-use integrity test passed on every run. The corrective action was to add the flux verification result to the batch record as a recorded attribute, so that a permeability trend across future lots can be detected before it becomes a performance issue.

Appendix A — Screening design matrix and responses

The coded design and the measured responses of the screening study are given in Table 13.2, and the same runs in natural units in Table 13.3. Factor letters are as defined in Table 4.2.

Table 13.2: Screening design in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
1	factorial	-1	-1	6.34	6.33	0.985
2	factorial	-1	1	6.08	6.41	0.972
3	factorial	1	-1	4.71	6.06	0.972
4	factorial	1	1	4.48	6.51	0.996
5	center	0	0	5.33	6.46	0.985
6	center	0	0	5.01	6.17	0.994
7	center	0	0	5.97	6.28	0.979

Table 13.3: Screening design in natural units, with the measured responses.

Run	Type	Filtration volume (load)	Filtration pressure	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
1	factorial	50	8	6.34	6.33	0.985
2	factorial	50	30	6.08	6.41	0.972
3	factorial	140	8	4.71	6.06	0.972
4	factorial	140	30	4.48	6.51	0.996
5	center	95	19	5.33	6.46	0.985
6	center	95	19	5.01	6.17	0.994
7	center	95	19	5.97	6.28	0.979

Appendix B — Response-surface design matrix and responses

The face-centred central composite design is given in coded factors in Table 13.4 and in natural units in Table 13.5.

Table 13.4: Response-surface design in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF (log_{10})	XMuLV LRF (log_{10})	Step yield
1	factorial	-1	-1	6.23	6.2	0.989
2	factorial	-1	1	6	6.34	0.985
3	factorial	1	-1	4.69	5.9	0.977
4	factorial	1	1	4.57	6.4	0.981
5	axial	-1	0	6.74	6.25	0.981
6	axial	1	0	4.51	6.38	0.979
7	axial	0	-1	5.11	6.15	0.975
8	axial	0	1	5.27	6.28	0.985
9	center	0	0	5.48	5.94	0.99
10	center	0	0	5.56	6.29	0.989
11	center	0	0	5.69	6.03	0.972
12	center	0	0	5.42	6.27	0.982

Table 13.5: Response-surface design in natural units, with the measured responses.

Run	Type	Filtration volume (load)	Filtration pressure	MVM LRF (log_{10})	XMuLV LRF (log_{10})	Step yield
1	facto-rial	50	8	6.23	6.2	0.989
2	facto-rial	50	30	6	6.34	0.985
3	facto-rial	140	8	4.69	5.9	0.977
4	facto-rial	140	30	4.57	6.4	0.981
5	axial	50	19	6.74	6.25	0.981
6	axial	140	19	4.51	6.38	0.979
7	axial	95	8	5.11	6.15	0.975
8	axial	95	30	5.27	6.28	0.985

Run	Type	Filtration volume (load)	Filtration pressure	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
9	center	95	19	5.48	5.94	0.99
10	center	95	19	5.56	6.29	0.989
11	center	95	19	5.69	6.03	0.972
12	center	95	19	5.42	6.27	0.982

Appendix C — Full effect and coefficient tables

Screening effect estimates for the two responses not shown in §5.2 are given in Table 13.6 and Table 13.7. Response-surface coefficients and the analysis of variance for the same two responses follow in Table 13.8 to Table 13.11, and the diagnostics for the XMuLV model in Figure 13.1.

Table 13.6: Screening effect estimates for the XMuLV log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.262	0.131	0.0608	2.16	0.12	
AB	0.188	0.0939	0.0608	1.54	0.22	
A	-0.0821	-0.041	0.0608	-0.675	0.548	

Table 13.7: Screening effect estimates for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	0.0181	0.00905	0.00365	2.48	0.0892	
A	0.00567	0.00283	0.00365	0.776	0.494	
B	0.00539	0.00269	0.00365	0.738	0.514	

Table 13.8: Response-surface coefficients for the XMuLV log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.16	0.0719	85.7	1.7e-10	***
A	-0.0191	0.0643	-0.297	0.776	
B	0.128	0.0643	1.99	0.0943	
AB	0.0896	0.0788	1.14	0.299	
A^2	0.0849	0.0965	0.88	0.413	
B^2	-0.00707	0.0965	-0.0733	0.944	

Table 13.9: Response-surface coefficients for step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.982	0.00313	314	7.02e-14	***

Term	Coef.	Std. err.	t	p-value	Sig.
A	-0.00336	0.0028	-1.2	0.275	
B	0.0016	0.0028	0.573	0.587	
AB	0.00199	0.00342	0.58	0.583	
A^2	-0.000294	0.00419	-0.0701	0.946	
B^2	0.000228	0.00419	0.0544	0.958	

Table 13.10: Analysis of variance for the XMuLV response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.153	5	0.0306	1.23	0.398
Residual	0.149	6	0.0248	nan	nan
Lack of fit	0.0562	3	0.0187	0.606	0.655
Pure error	0.0927	3	0.0309	nan	nan

Table 13.11: Analysis of variance for the step-yield response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.9e-05	5	1.98e-05	0.422	0.819
Residual	0.000282	6	4.69e-05	nan	nan
Lack of fit	7.02e-05	3	2.34e-05	0.332	0.805
Pure error	0.000211	3	7.05e-05	nan	nan

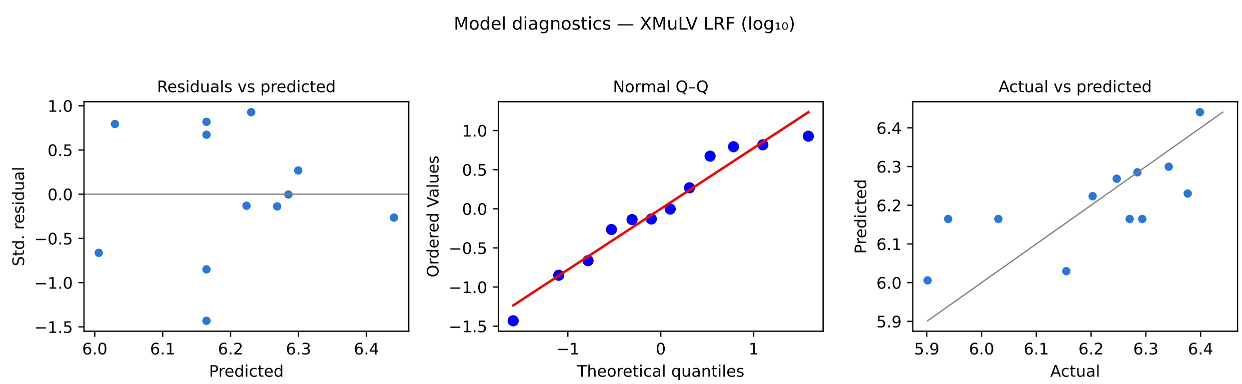

Figure 13.1: Model diagnostics for the XMuLV log reduction factor. The actual-against-predicted panel shows the narrow range of prediction that follows from a response with no resolved factor effect.

Appendix D — Analytical methods summary

Performance of the two validated infectivity assays used by the step is summarized in Table 13.12. The assay share of variance is the fraction of the observed run-to-run variance in the corresponding response that is attributable to the method.

Table 13.12: Validated analytical methods for the step, with reported precision, limit of quantitation, accuracy and the assay share of the observed variance.

Reference	Method	Precision ($\log_{10}$)	LOQ ($\log_{10}$)	Accuracy (%)	Assay share of variance
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32	1.4	96	0.3
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5	96	0.28

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